رواد مصر الرقمية

وزارة الاتصالات
وتكنولوجيا المعلومات

Loan Risk Detection Project

Team Members :

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Project Description:

The objective of this project is to develop a machine learning system capable of predicting loan default risk for financial institutions. Loan risk detection is a critical task in the finance sector, as it helps organizations evaluate the likelihood of a borrower failing to repay a loan, enabling better decision-making and reducing financial losses.

This project focuses on analyzing borrower data such as demographics, income, credit history, employment status, and loan details to identify patterns associated with default risk.

The workflow includes data collection, preprocessing, exploratory data analysis (EDA), feature engineering, model development, and evaluation. Multiple machine learning algorithms will be applied and compared to select the most effective model.

Dataset :

From Kaggle dataset

About Dataset :

Banks earn a major revenue from lending loans. But it is often associated with risk. The borrower's may default on the loan. To mitigate this issue, the banks have decided to use Machine Learning to overcome this issue. They have collected past data on the loan borrowers & would like you to develop a strong ML Model to classify if any new borrower is likely to default or not.

Description :

The dataset is enormous & consists of multiple deteministic factors like borrowe's income, gender, loan pupose etc. The dataset is subject to strong multicollinearity & empty values. Can you overcome these factors & build a strong classifier to predict defaulters ?

Methodology:

1- Data Collection & Preprocessing

- Handle missing values
- Encode categorical variables
- Normalize numerical features

2- Exploratory Data Analysis (EDA)

- Identify trends and correlations
- Detect key factors influencing loan default

3- Feature Engineering

- Create new meaningful features
- Select important variables

4- Model Development

- Logistic Regression
- Decision Trees
- Random Forest
- Support Vector Machines (SVM)

5- Model Evaluation

- Accuracy
- Precision
- Recall
- F1 Score
- ROC-AUC

6- Deployment

- Simple web interface or dashboard for real-time predictions

Expected Deliverables:

- Cleaned and processed dataset
- Trained machine learning models
- Interactive dashboard or web interface
- Final report with insights and recommendations

Tools and Technologies:

- Python
- scikit-learn
- pandas
- NumPy
- Power BI / Streamlit (for visualization and deployment)

Project Management & Execution Plan

Project Timeline (8 Weeks):

- **Week 1:** Understanding loan risk & project requirements
- **Week 2–3:** Data collection & preprocessing
- **Week 4–5:** EDA & feature engineering
- **Week 6–7:** Model development & evaluation
- **Week 8:** Deployment + report & presentation

Milestones:

- Dataset prepared
- Insights extracted from EDA
- Models trained and evaluated
- Final model selected
- Dashboard/interface completed
- Final report delivered

Task Assignment & Team Roles :

1. **Project Manager / Business Analyst**
    - Define objectives and KPIs
    - Manage workflow and coordination
2. **Data Engineer**
    - Data collection and preprocessing
3. **Data Analyst**
    - Perform EDA and extract insights
4. **Machine Learning Engineer**
    - Build, train, and evaluate models
5. **Visualization & Deployment Specialist**
    - Create dashboard / deploy model
    - Prepare final presentation

Risk Assessment & Mitigation Plan:

Risk	Impact	Mitigation
Poor data quality	Low accuracy	Data cleaning & preprocessing
Imbalanced data	Biased predictions	Use SMOTE / resampling
Overfitting	Poor generalization	Cross-validation & regularization
Time constraints	Delay	Proper task distribution
Deployment issues	System failure	Early testing & simple tools

KPIs (Key Performance Indicators):

Model Performance:

- Accuracy
- Precision
- Recall
- F1 Score
- ROC-AUC

System Performance:

- Response time
- Reliability

Business Impact:

- Improved risk detection
- Reduction in loan defaults

Usability:

- Ease of use
- Clarity of insights

Project Outcome:

This project will deliver a practical solution that enables financial institutions to **identify high-risk applicants,** improve lending decisions, and minimize potential losses through data-driven insights.